MADHULIKA VIKRAMAN

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EDUCATION

University of Southern California — **Los Angeles, California**
Master of Science in Computer Science – 3.8/4.0 GPA — **August 2025-May 2027**
Courses-Deep Learning and its Applications, Machine Learning with Data Science, Analysis of Algorithms, Database Systems, Web Technologies

PES University, RR Campus — **Bangalore, India**
Bachelor of Technology in Computer Science - 9.29/10 GPA, 4-semester merit scholarship recipient — **December 2020-July 2024**

WORK EXPERIENCE

University of Southern California - Norman Lear Center (Ongoing) — Los Angeles, CA
Student Research Engineer — May 2026 - Present

- Conducted **LLM-powered sentiment analysis** of Reddit discussions on public discourse surrounding clinical trials in Grey's Anatomy using a research codebook with human validation and comparison.
- Built a **Python NLP pipeline (Ollama-served LLMs)** that automated qualitative coding, statistical analysis, inter-rater reliability, and data visualization and produced visualizations to support health communication research at the **USC CTSI**.

Celestica — Chennai, India
Associate Software Engineer — September 2024-July 2025

- Owned the end-to-end software development of multiple SONiC community **CLI** features, using **Python, Go, YANG, Click/Klish**.
- Resolved critical bugs across releases, reducing release-time issues by 30%
- Automation - **Ansible** to generate and execute state-changing device configs, improving efficiency by ~ 50%, reducing manual intervention

Sandvine Technologies (Applogic Networks) — Bangalore, India
Software Development Intern — January 2024-August 2024

- Rebuilt a data conversion/transport feature, in **Python**, along with simultaneous data collection and analysis, optimizing it by over 60%
- Built modules for an **RDBMS**-based network traffic classification system within the Active Network Intelligence Engine (**Python, SQL, REST API, Data analysis**)
- Built client-server system in **Go**, to simulate real-time traffic (custom structures from parquet files) to server(s) over a **TCP** connection

SKILLS

Programming languages & Databases: Python, Go, C, C++, Bash, Yang, R, Java, SQL, MySQL, MongoDB, Redis
AI/ML: PyTorch, TensorFlow, HuggingFace, Keras, Scikit-learn, LLMs, LangChain, LangGraph, RAG, GenAI, Agentic AI
DevOps: Docker, Git, Ansible, Azure, Jenkins
Full-stack Development: HTML5, CSS, JavaScript, Flask, FastAPI
Coursework: Data Structures, Artificial Intelligence (AI), Machine Learning (ML), Cloud Computing (AWS EC2, S3), Deep Learning, Natural Language Processing (NLP), Algorithms, Computer Networks, Data Analytics, Big Data, Operating Systems, Statistics, Web technologies, Databases, Research Methodology, Reinforcement Learning, Time Series Modeling, Multimodal AI

ACADEMIC PROJECTS

- **Legal LLM [Ongoing, USC GRIDS Club]:** LLM-based system to identify risky/predatory clauses in Rental/Lease agreements, implementing RAG pipelines using FAISS and different LLMs to generate context-aware answers from parsed documents.
- **Heart disease and Seizure Prediction using Liquid Neural Networks and Neural Differential Equations** (Research project for Deep Learning)**:** Explored the comparison of traditional benchmark deep learning methods and Liquid Time Constant Networks for multivariate time series prediction of Heart-Disease and Seizures from EEG/EKG signal data, preparing for publication.
- **Sarcasm, Irony, and Humor Detection in Low-Resource Languages (Capstone project, Team Size: 4) - NLP, ML, DL, PyTorch, TensorFlow, Keras** : Implemented existing and custom ensemble Machine Learning and Deep Learning models (30 total) for text classification in Kannada and Tulu. Pioneered an early dataset in Tulu. Achieved 86-89% accuracy in classification. Presented the research paper at ICICV 2024 conference, later published in Springer Lecture Notes in Networks and Systems.
- **Gymbot - GenAI, Python, SQL** : Programmed a chat assistant leveraging Google's GenAI tools to help users log, modify, and query daily fitness data stored in a database. Integrated database operations and model-driven responses for logged data information and inferences, fitness and dietary advice for the **Google GenAI hackathon** hosted by **Kaggle.**
- **Crop Yield Prediction - Python, ML, DL, Streamlit, Docker** : Developed a containerized web interface for predictive modeling of public agricultural yield data (district-wise) and selection of features to forecast crop yield, aiding data-driven decision-making.
- **Real Estate Price Prediction - Python, Data Visualization** : Conducted data analysis on Zillow Home Prices dataset and developed predictive models using Python to forecast house prices and optimize ROI for investors.
- **Third Eye for the Blind - Arduino, C, TinkerCAD** : Designed a portable obstacle detection system using an Arduino microcontroller, ultrasonic sensors, and a buzzer to assist visually impaired individuals with navigation.
- **Quiz System - Java Spring Boot, MySQL** : Built a system with access control for 4 roles, quiz creation, management, performance tracking
- **Home Assistant & Security System - Python** : Created a voice-activated assistant capable of basic command recognition and home surveillance through image detection and facial recognition.